

**Table I: Specifications of Some Smart Tweezers**

| Smart Tweezers                 | INDE Enterprises ST-5R                                      | Advance Tech ST-5L                                                                     | Advance Devices ST5                                         | IDEAL - TEK ST-1R                                                    |
|--------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------|----------------------------------------------------------------------|
| <b>Basic specifications</b>    |                                                             |                                                                                        |                                                             |                                                                      |
| Measured parameter             | C, L, R, ESR, Rs, Rp                                        | C, L, R, ESR, Rs, Rp                                                                   | L, C, R, Z, ESR                                             | C+R (ESR), L+R, R                                                    |
| Measuring frequency            | 100 Hz, 120 Hz, 1 kHz, 10 kHz                               | 100 Hz, 120 Hz, 1 kHz, 10 kHz                                                          | 100 Hz, 120 Hz, 1 kHz, 10 kHz                               | 100 Hz, 1 kHz, 10 kHz                                                |
| Measurement rate               | 1 time per second, default                                  | 1 time per second, default                                                             |                                                             | 4, 2, 1, 0.5 times per second (default 1)                            |
| DC voltage                     | No                                                          | 8 volts +/-                                                                            |                                                             | 0 to 800 mV (up to 8V with manual setting)                           |
| Resistance range               | 0.05 Ohms to 9.99 MOhms                                     | 0.05 ohms to 9.99 mega-ohms                                                            | 0.05 $\Omega$ to 9.9 M $\Omega$                             | 0.1 to 5 mega-ohms                                                   |
| Capacitance range              | 0.5 pF to 4999 $\mu$ F                                      | 0.5 pF to 4999 $\mu$ F                                                                 | 0.5 pF to 4999 $\mu$ F                                      | 1 pF to 4999 $\mu$ F (1 pF - 5 pF in manual mode)                    |
| Inductance range               | 1 $\mu$ H to 999 mH                                         | 1 $\mu$ H to 999 mH                                                                    | 0.5 $\mu$ H to 999 mH                                       | 1 $\mu$ H to 499 mH (1 $\mu$ H -4 $\mu$ H in manual mode)            |
| <b>Accuracy specifications</b> |                                                             |                                                                                        |                                                             |                                                                      |
| Resistance Range               | 0.1 ohm to 5 mega-ohms                                      | 0.1 ohm to 5 mega-ohms                                                                 | 0.05 ohm to 9.9 mega-ohms                                   | 10 ohms to 5 mega-ohms                                               |
| Accuracy                       | 0.2% in range 0.1R - 5M                                     | 0.2% in range 0.1R - 5M                                                                | Up to 0.2%                                                  | 1% + 0.02 in the range 10 ohms - 5 mega-ohms                         |
| Test frequency                 | 1 kHz                                                       | 1 kHz                                                                                  | 1 kHz                                                       | 1 kHz                                                                |
| Capacitance Range              | 0.5 pF to 4999 $\mu$ F                                      | 0.5 pF to 4999 $\mu$ F                                                                 | 0.5 pF to 4999 $\mu$ F                                      | 20 pF to 4999 $\mu$ F                                                |
| Accuracy                       | 0.5% in range 10 pF - 499 $\mu$ F                           | 0.5% in range 10 pF - 499 $\mu$ F (resolution: 0.5 pF in range 1 pF to 100 pF)         | Up to 0.2%                                                  | 3% + 1.7 pF; typical offset 1.6 pF for 0402 size component           |
| Test frequency                 | 1 kHz C > 1000 pF; 10 kHz C < 1000 pF; 100 Hz C > 1 $\mu$ F | 1 kHz C > 1000 pF; 10 kHz C < 1000 pF; 100Hz C > 1 $\mu$ F                             | 1 kHz C > 1000 pF; 10 kHz C < 1000 pF; 100 Hz C > 1 $\mu$ F | 1kHz C >1000 pF, 10 kHz C <1000 pF, 100 Hz C >1 $\mu$ F              |
| Inductance Range               | 1 $\mu$ H to 1H                                             | 1 $\mu$ H to 1 H                                                                       | 0.5 $\mu$ H to 999 mH                                       | Range: 10 $\mu$ H to 499 mH                                          |
| Accuracy                       | 0.5% in range 1 $\mu$ H - 1H                                | 0.5% in range 1 $\mu$ H - 1H (resolution 0.5 $\mu$ H in range 1 $\mu$ H - 100 $\mu$ H) | Up to 0.5%                                                  | 3% + 0.4 $\mu$ H; Typical offset 1.4 $\mu$ H for 0402 size component |
| Test frequency                 | 10 kHz L < 1 $\mu$ H; 1 kHz L > 1 $\mu$ H; 100Hz L > 1 mH   | 10 kHz L < 1 $\mu$ H; 1 kHz L > 1 $\mu$ H; 100 Hz L > 1 mH                             | 10 kHz L < 1 $\mu$ H; 1 kHz L > 1 $\mu$ H; 100 Hz L > 1 mH  | 10 kHz L <1 mH, 1 kHz L >1 mH, 100 Hz L <10 mH                       |
| <b>Physical specifications</b> |                                                             |                                                                                        |                                                             |                                                                      |
| Operating temperature          | 0°C to 50°C                                                 | 0°C to +55°C                                                                           | 0°C to +55°C                                                | 0°C to +55°C                                                         |
| Battery type                   | 3.7V LiPO rechargeable 150mAh                               | Rechargeable 4.5V Li-Ion battery                                                       | 3.7V LiPO rechargeable 150mAh                               | 3x1.5V LR44 alkaline or air zinc                                     |
| Battery life                   | 80 hours for fully charged battery                          | 80 hours for fully charged battery                                                     | 80 hours typical, 2 hours charging cycle                    | 80 hours for alkaline, 240 hours for zinc air                        |
| Size                           | 14.0x2.5x3.0 cm3                                            | 14.0x2.5x3.0 cm3 (3.94x0.9x1.5 inches3)                                                | 14.0x2.5x3.0 cm3 (3.94x0.9x1.5 inches3)                     | 14.0x2.5x3.0 cm3 (3.94 x 0.9 x 1.5 inches3)                          |
| Weight                         | 53 grams                                                    | 53 grams                                                                               | 53 grams                                                    | 53 grams                                                             |